

Lead Finder AI Automation

A hyperlocal lead generation and CRM automation platform — scraping, managing, and visualizing business intelligence from Google Maps, JustDial & IndiaMart.

Next.js 15

Node.js + Express

MongoDB

Playwright

TypeScript

Recharts

React Leaflet

Zustand

3

Scraping Sources

5

CRM Pipeline Stages

8

App Modules

Platform — Lead Finder AI Automation
Type — Full-Stack SaaS · Production Grade
Stack — Next.js · Node.js · MongoDB · Playwright

Table of Contents

01	Project Overview
02	Project Objectives
03	Key Features
04	Technology Stack
05	Project Architecture
06	Search & Scraping System
07	Lead Management System
08	CRM Pipeline System
09	Automation System
10	Analytics Dashboard
11	Area Heatmap
12	Database Design
13	Security Features
14	Performance Optimizations
15	Final Summary
16	Project Screenshots

01 Project Overview

Lead Finder AI Automation is a **production-grade, AI-powered lead generation and CRM automation platform** designed for modern business intelligence and sales workflows. The platform automatically discovers, scrapes, organizes, analyzes, and manages business leads from multiple online sources.

The system uses a geographic hierarchy — **State** → **City** → **Area** — to enable hyperlocal lead discovery, making it the ideal tool for sales teams operating in specific regions or localities.

Data Sources

- ▶ Google Maps
- ▶ JustDial
- ▶ IndiaMart

Target Audience

- ▶ Lead generation agencies
- ▶ Sales teams
- ▶ Marketing companies
- ▶ CRM operations
- ▶ Business intelligence workflows

02 Project Objectives

The main objective is to **automate hyperlocal business lead generation** with real-time CRM management and analytics. The platform provides a unified ecosystem that bridges the gap between raw data discovery and actionable sales intelligence.

 Multi-Source Business Scraping

 Real-time Lead Storage

 CRM Pipeline Workflows

 Automation Systems

 Area-wise Analytics

 Export Systems

 Business Intelligence Dashboards

03 Key Features

Multi-Source Scraping

- ▶ Google Maps integration
- ▶ JustDial data extraction
- ▶ IndiaMart business discovery

Real-time Lead Management

- ▶ Instant MongoDB storage
- ▶ Dynamic frontend updates
- ▶ Real-time synchronization

CRM Pipeline

- ▶ New Lead → Contacted → Interested
- ▶ Follow-Up tracking
- ▶ Not Interested management

Automation Engine

- ▶ State-wise auto scraping
- ▶ City-wise scraping
- ▶ Area-wise multi-business search

Analytics Dashboard

- ▶ Total leads & searches
- ▶ Area heatmaps
- ▶ Business density reports
- ▶ Automation progress

Export & Filters

- ▶ CSV & Excel export
- ▶ Category / Source filters
- ▶ Website & phone filters
- ▶ Dynamic pagination

04 Technology Stack

Frontend Technologies

TECHNOLOGY	PURPOSE
Next.js 15	Frontend Framework
React 19	UI Library
TypeScript	Type Safety
Tailwind CSS, MUI.com	Styling
Zustand	State Management
React Query	API Caching
Axios	API Requests

Backend Technologies

TECHNOLOGY	PURPOSE
Node.js	Runtime Environment
Express.js	Backend REST APIs
TypeScript	Backend Type Safety
MongoDB	Primary Database
Mongoose	ODM (Object Data Modelling)
Playwright	Browser Scraping Engine
Chromium	Headless Browser Automation

05 Project Architecture

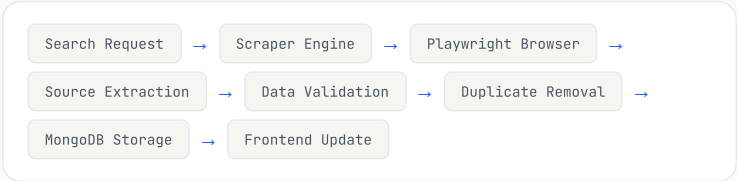
Frontend — Modular Component-Based Architecture

MAIN PAGES	BACKEND MODULES	DATABASE COLLECTIONS
- Dashboard - Search	- Authentication - Scraper Engine	- users - leads

- Leads
 - CRM Pipeline
 - Analytics
 - Automation
 - Settings
- Lead Management
 - CRM Services
 - Automation Engine
 - Analytics Engine
 - Export Manager
- crmPipeline
 - automationJobs
 - analytics

06 Search & Scraping System

The user inputs a **Business Type**, **State**, **City**, **Area** and selects sources. Example: [Salon](#) [Gujarat](#) [Ahmedabad](#) [Naroda](#). The Playwright scraping engine then processes this through a multi-step validation pipeline.



Extracted Business Data Fields

- Business Name
- Website
- Phone Number
- Address
- Rating
- Reviews
- Business Category
- Source Platform

07 Lead Management System

The Leads page is the central hub for viewing and managing all collected business leads with real-time updates, intelligent filtering, and responsive card layouts. Latest leads always appear first.

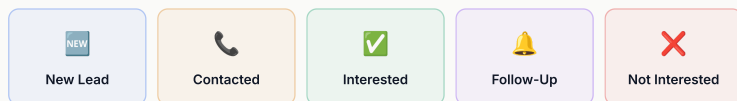
Available Filters	Page Features
- Category - Source Platform	- Real-time lead updates - Dynamic filtering

- ▶ Area
- ▶ Website Available
- ▶ Phone Number Available

- ▶ Pagination (Prev / Current / Next)
- ▶ Responsive lead cards
- ▶ CSV & Excel export options

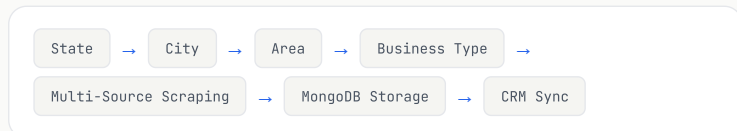
08 CRM Pipeline System

The CRM system enables structured, stage-based lead management. Each lead flows through a visual pipeline with dynamic status updates, follow-up tracking, notes management, and priority assignment — all synchronized in real-time.



09 Automation System

The Automation Engine automates area-wise business scraping across multiple categories simultaneously. It supports resume capability — if automation stops mid-process, it resumes from the last completed position, preserving all previously completed jobs.



Multi-Business Automation

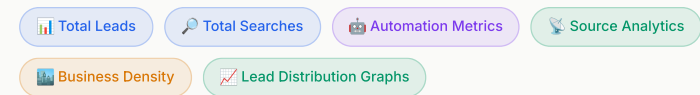
- ▶ Salon, Gym, Grocery Store
- ▶ Auto-creates scraping jobs per category
- ▶ Per area combination support

Resume Automation

- ▶ Resume from last position
- ▶ Continue pending jobs
- ▶ Preserve completed jobs

10 Analytics Dashboard

The Analytics page provides real-time business intelligence insights with dynamic charts, distribution graphs, and source-specific analytics — giving teams full visibility into lead quality and scraping performance.



11 Area Heatmap

The interactive heatmap visualizes business density geographically across areas. Dynamic markers with area tooltips allow users to click and filter leads by region instantly. Example:

Satellite → 420 Businesses · Prahlad Nagar → 310 · Naroda → 120



12 Database Design

COLLECTION	PURPOSE
users	User authentication and session management
leads	All scraped business lead records
crmPipeline	CRM stage tracking and workflow data
automationJobs	Automation job queue and progress tracking

COLLECTION	PURPOSE
analytics	Reporting metrics and aggregate statistics

13 Security Features

 JWT Authentication

 Protected Routes

 Session Persistence

 Admin-Only Access

 Secure API Middleware

14 Performance Optimizations

Backend Optimizations

- Queue-based scraping
- Browser reuse across sessions
- Retry handling for failed requests
- Parallel multi-source extraction
- Batched MongoDB writes

Frontend Optimizations

- Pagination for large datasets
- Lazy rendering components
- React Query cached APIs
- Real-time sync with WS/polling

15 Final Summary

Enterprise-Grade Lead Intelligence Platform

Lead Finder AI Automation is a scalable, production-ready ecosystem capable of hyperlocal business scraping, CRM workflow automation, real-time analytics, area-wise automation, multi-source lead discovery, and business intelligence visualization.

Designed for sales agencies, marketing companies, business intelligence teams, and automated lead generation operations — providing a modern SaaS-style architecture optimized for scalability, real-time workflows, and production-grade performance.

HyperLocal Scraping

CRM Automation

Real-time Analytics

Area Heatmaps

Multi-source Discovery

BI Visualization

JWT Security

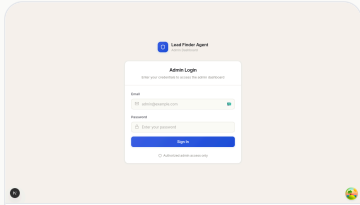
Export Ready

Project Screenshots

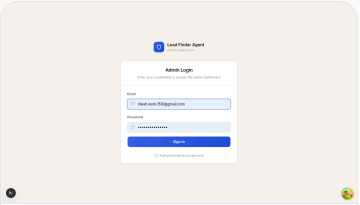
Visual walkthrough of all major modules and screens

1) Login Page

Authentication



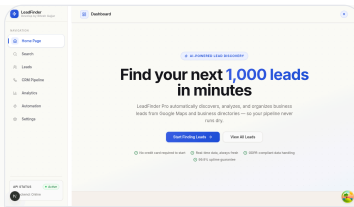
Login Page · I1.png



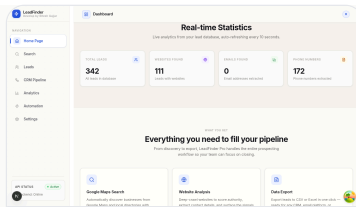
Login Page · I2.png

2) Dashboard

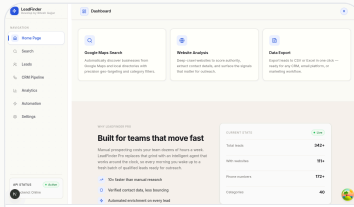
Overview & Stats



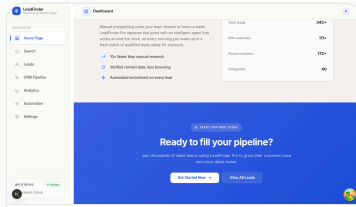
Dashboard • I1.png



Dashboard • I2.png



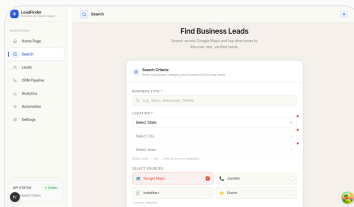
Dashboard • I3.png



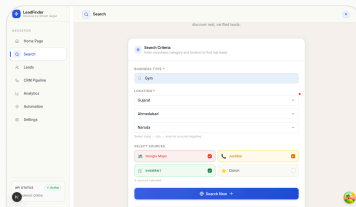
Dashboard • I4.png

3) Search

Scraping Engine



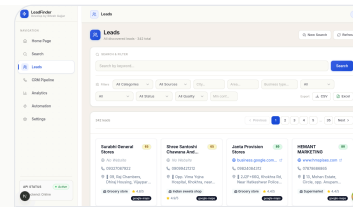
Search • I1.png



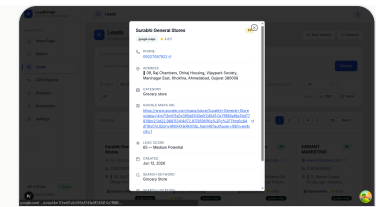
Search • I2.png

4) Leads

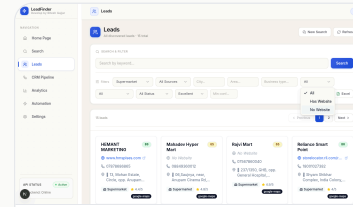
Lead Management



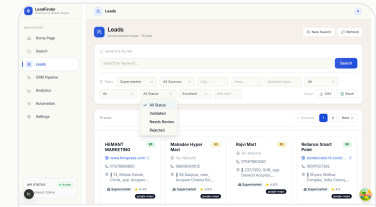
Leads • I1.png



Leads • I2.png



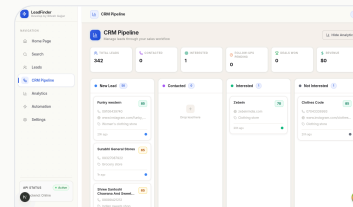
Leads • I3.png



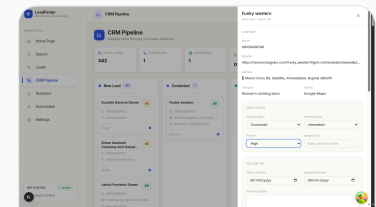
Leads • I4.png

5) CRM Pipeline

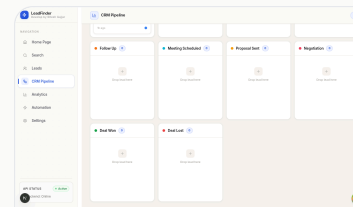
Workflow Stages



CRM Pipeline • I1.png



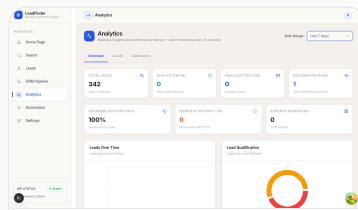
CRM Pipeline • I2.png



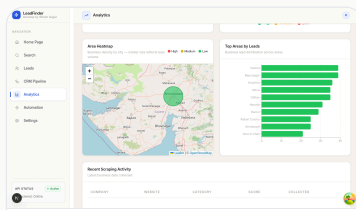
CRM Pipeline • I3.png

6) Analytics

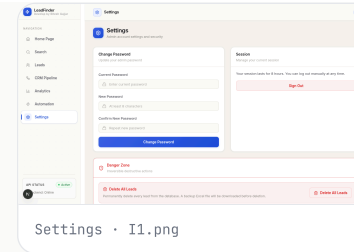
Business Intelligence



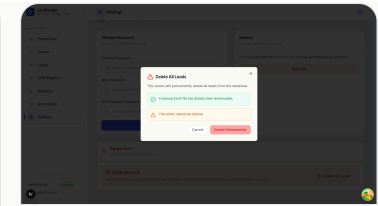
Analytics · I1.png



Analytics · I2.png



Settings · I1.png

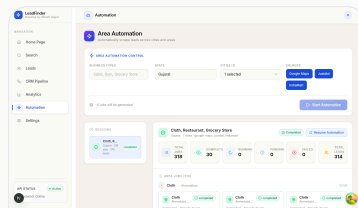


Settings · I2.png

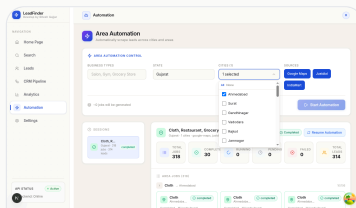
Lead Finder AI Automation – Project Documentation · Ritesh Gajjar

7) Automation

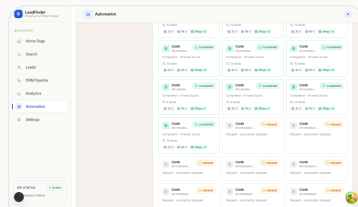
Auto Scraping Engine



Automation · I1.png



Automation · I2.png



Automation · I3.png

8) Settings

Configuration